

Mohammed Haitham El Zamzamy

AI Engineer | LLMs and RAG, Computer Vision, Full-Stack, Prompt Engineering, MLOps

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PROFESSIONAL SUMMARY

AI Engineer and full-stack builder with **two IEEE-published deep-learning papers** and a **national AI Olympiad medal (2nd place)**. As the **first and sole engineer** at a Saudi ride-hailing operator, designed, built, and operate its entire internal software stack, **roughly 15 production systems** spanning customer support, CRM, AI driver-screening, incentives, dispatch, and internal communications, while **coordinating and mentoring a 3-person operations and content team**. Deep across **LLMs and RAG, computer vision, and production full-stack engineering** (TypeScript, Python, React, Next.js, PostgreSQL), with a test-heavy, ship-to-production discipline.

TECHNICAL SKILLS

AI and Machine Learning: LLMs (Gemini, GPT-4o, OpenAI, Claude), RAG, prompt engineering, vision-language models (VLM) and OCR, PyTorch, TensorFlow, scikit-learn, CNNs, U-Net and Attention U-Net, EfficientNet, LSTM, transfer learning, model evaluation, OpenCV, YOLOv8

Languages: Python, TypeScript, JavaScript (Node.js), SQL, C++

Backend: Node.js, Express, Next.js (App Router, Server Actions), Drizzle ORM, Prisma, Zod, REST and OpenAPI, JWT and session auth

Frontend: React, Next.js, Vite, Tailwind CSS, shadcn/ui, TanStack Query, Streamlit

Databases and Data: PostgreSQL (Neon), SQLite, SQL migrations, triggers, append-only ledgers, advisory locks, Pandas, NumPy, feature engineering

Cloud and DevOps: AWS (SageMaker, S3), Vercel, DigitalOcean (droplets, nginx, systemd, TLS), Replit, Docker, GitHub CI/CD, Kafka, node-cron

Integrations and Automation: n8n, Meta WhatsApp Cloud API, Google Maps, LiveKit and WebRTC, Uber H3, Gmail/IMAP, ONDE and WASL/TGA APIs

Testing: Vitest, Playwright, fast-check (property-based), pytest

EXPERIENCE

AI and Full-Stack Software Engineer, SHEGO

Remote, Saudi Arabia | Jun 2026 - Present

Women-only ride-hailing operator running three regional companies on the ONDE platform. First and sole engineer; own the entire internal software stack end to end.

- Designed, built, and operate **roughly 15 Arabic-first production systems**, owning architecture, database migrations, deployment, and live incident response, and **coordinate and mentor a 3-person operations and content team**, reviewing their work and helping them improve.
- Built and operate the flagship **customer-support platform** (help desk plus CRM): **195 PostgreSQL tables, 122 React pages, 91 API modules**, roughly **7,500 live tickets** from WhatsApp and email, an advisory-lock **auto-assignment and routing engine**, SLA escalation, an **84-toggle role-based access control** model, and a **27-section quality-evaluation system**, gated by **2,300-plus automated tests** on every deploy. **React, Next.js, Express, PostgreSQL on Neon.**
- Built an **AI driver-onboarding and compliance-screening system: vision-language document extraction (Gemini and GPT-4o)** from Saudi licenses, national IDs, and iqamas, Arabic-name gender fusion, and a **WASL/TGA compliance decision engine** verified row-for-row against the official Saudi document. Backed by **1,664 automated tests**, remediated roughly **5,000 drivers**, and caught real identity fraud.
- Shipped an **AI and OCR financial-verification pipeline** (reads IBAN, amount, and beneficiary from documents with a GPT-4o vision model, matched deterministically) and a **confidence-gated AI auto-reply** system with customer memory and a usage-cost ledger.
- Built a **driver incentive and retention system** (Next.js, Prisma, Neon) with a money-critical **append-only ledger** and **serializable transactions enforcing trip caps under concurrency**, plus a Gemini-driven **smart-surge engine** issuing real driver payouts.
- Built a **demand-forecasting and dispatch system** (Poisson modeling, **Uber H3** zones, forecast-accuracy A/B ledger, Google Maps console) and an automated **female-driver leads pipeline** ingesting Meta, TikTok, and Snapchat lead ads to send WhatsApp recruitment templates.
- Built an **internal team-communication platform** (custom Slack/Teams) with real-time **LiveKit audio and video for up to 100 participants**; plus a **CRM replacing Zoho** (27 features on n8n), a request-lifecycle tracker (Postgres triggers, Gemini subtask splitter), a tasks and approval-chains system, a referral workspace, and a training-quiz platform.
- Own all production operations: **160-plus database migrations**, deployments (Replit, Vercel, DigitalOcean), API key rotations, resolving full production outages, and **live as-a-user verification (Playwright) before every release**.

AI Engineer (Healthcare), GraviLog

Remote, Contract | Oct 2025 - Present

- Built a fetal-ultrasound **segmentation model (Attention U-Net with EfficientNet-B3)** reaching **0.75 mean Dice** across **1,588 images from 169 patients**, with patient-level splits, custom augmentation, and mixed-precision training.
- Delivered an end-to-end clinical inference pipeline (Grad-CAM explainability, structured measurement extraction) focused on reproducibility and clinically usable output.

AI Engineer, EVUNITY

On-site, Cairo | Nov 2025 - Sep 2026

- Build **AI-driven automation workflows and web-based AI tools**, integrating LLM components, OAuth-based API connections, and intelligent backends into production systems, from prompt design through deployment.
- Ship internal and client-facing AI products end to end, combining web development and automation across product and engineering teams.
- Stood up **AABAR Mentor** (mentor.aabar.io) as a separate **Go and React** deployment on a DigitalOcean droplet with its own database and preserved SSO and GPT-4 flow, live over HTTPS.

AI Engineer (Automation), Exavo.ai

Remote, Contract | Sep 2025 - Jun 2026

- Contributed to **7 production AI services**, including AI audio transcription, email auto-reply drafting, social-media automation, custom AI agents, AI website builders, Shopify product automation, and smart visual insights, using **n8n, Replit, and Lovable**.
- Built and deployed **n8n automation workflows** for daily content publishing, integrating the Google, LinkedIn, and OpenAI APIs end to end, resolving OAuth and credential issues.
- Developed reusable **AI agents and prompt flows in Microsoft Copilot Studio** for internal team operations.

AI Engineering Intern, Algoriza / DMC-CURVE / Slash Hub

Remote and Hybrid | 2024 - 2025

- **Algoriza**: contributed to ML and LLM prototypes and model evaluation in Python (PyTorch, scikit-learn, pandas), built data-preprocessing pipelines, and wrote clean, peer-reviewed code.
- **DMC-CURVE**: built **LLM and RAG** proofs-of-concept integrating vector stores with conversational agents, with prompt and agent design for multi-turn flows.
- **Slash Hub**: engineered visual features from product-image datasets and trained **CNN models** for product image recognition and a visual-similarity recommender.

PROJECTS

- **Heavy-Equipment Computer Vision Pipeline**: real-time equipment activity tracking with YOLOv8 detection, a custom hybrid tracker, masked optical flow, Kafka streaming, PostgreSQL, a Streamlit dashboard, and Docker Compose.
- **Jet-Bridge and Boarding-Stairs Detection**: YOLOv8 segmentation detecting aircraft body type and jet-bridge and stairs connections from airport gate video, with a Groq LLaMA vision step.
- **Fine-Tuning BERT and GPT for Text Tasks**: transformer fine-tuning for sentiment and emotion detection using transfer learning and embedding extraction.
- **Ionospheric Forecasting, Egyptian Space Agency**: deep LSTM forecasting regional VTEC patterns and GNSS disruptions with a physics-informed loss, real-time GUI, and uncertainty estimates.
- **Multi-Sensor Fusion for Vehicle Distance Estimation**: Kalman filtering and signal integration across multiple sensor inputs.
- **Amazon Electronics Recommender**: collaborative-filtering recommendation engine on customer reviews and behavior.

PUBLICATIONS (IEEE) AND AWARDS

- **IEEE IMCOM 2026**: "Explainable Ensemble Deep Learning Architecture for Fruit Disease Detection for Sustainable Agriculture and Food System Resilience," **98.47 percent accuracy** with Grad-CAM explainability. [IEEE Xplore](#)
- **IEEE CSDGAIS 2024**: "A Novel Deep CNN Model with L1 Regularization for Plant Disease Classification."
- **AI Olympiad, 2nd Place Nationally** (Egypt, 2024), across Computer Vision, Embedded Systems, and Cybersecurity.

CERTIFICATIONS

Introduction to Transformer-Based NLP, **NVIDIA** (2025) | AWS Academy Machine Learning Foundations and Cloud Foundations, **AWS** | Advanced Learning Algorithms and Supervised Machine Learning, **DeepLearning.AI and Stanford** | Data Scientist Professional Certification, **DataCamp** | Deep Learning with PyTorch, DataCamp | Meta Front-End (React, JavaScript)

EDUCATION AND LANGUAGES

BSc in Artificial Intelligence, Galala University, Egypt, 2025. Grade: Very Good. Graduation project with the Egyptian Space Agency. | **Languages**: Arabic (native), English (professional). | Available immediately.